POST-2026-05-26 REWRITE BANNER (D-tier, binding 2026-05-26 EOD) This paper is a D-tier (REWRITE) entry per `Docs/policy/PAPERS_REVISION_PLAN_2026_05_26.md`. The original body is preserved as historical reference; the post-2026-05-26 reclassification re-frames the paper's claims as documented below. Reader: treat the original abstract / introduction / conclusion as DRAFT-stage content superseded by this banner.

Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* TOE-level claim SUSPENDED until non-standard rescue independently derived (per Math411-AddB §10).

Post-cascade re-framing for this paper: Paper-11 was titled "Pillar 11 Λ cancellation" but in the canonical numbering this content is actually **Pillar 8 cosmological constant suppression** (C1 core emergence, T7). The current Pillar 11 designation is *dark-sector phenomenology* (C3 phenomenological add-on). The paper's Λ-cancellation content is preserved as Pillar 8 (T7 unchanged) plus a separate (small) dark-sector section needing C3 re-framing.

What survives: The Math58-v7 four-sector cancellation chain (Λ suppression mechanism) is Pillar 8 content and stands at T7 PROVED. Group-theoretic and topological derivations unchanged.

What is downgraded: Any dark-sector phenomenology claims in this paper attached to "Pillar 11" are reframed under C3: TECT does NOT predict dark-matter from internal axioms; the Pillar 11.A wall-DM is T1 REFUTED, Pillar 11.A textures are T2 INTERIM NEGATIVE (compactness AUDIT), Pillar 11.B ν_R bulk is DOUBLY BLOCKED at TECT-natural (Math412-AddA G3-A + Math412-AddB G3-B).

Forward path: Paper-11 should be split (in a future revision) into: (a) Paper-8-Lambda-Cancellation (C1 Pillar 8, T7 PROVED, ready-to-publish); (b) Paper-11-Dark-Sector-C3 (phenomenological matching framework; Math409-AddD-AddC programme). This addendum documents the conceptual split without performing the actual file split.

See: Math58-v7 (Λ four-sector cancellation, T7); Math412-AddA + Math412-AddB (Pillar 11.B DOUBLY BLOCKED); Math409-AddD-AddC (Pillar 11.A wall-DM REFUTED).

A four-sector cancellation programme note for the cosmological constant in TECT

(NOT a Pillar-11 closure paper; the present draft fails its own pre-registered falsification gate as written — internal numerical / gate inconsistency flagged for

revision.)

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We outline a candidate four-sector cancellation programme for the cosmological constant within the Topological Energy Condensate Theory (TECT). The BCC topological condensate is decomposed into four candidate independent sectors: (i) the BCC order-parameter self-interaction energy, (ii) the monopole sector (captured by the topological Chern class c_2), (iii) the Dirac-fermion zero-mode sector, and (iv) the vortex sector (defect core energy).

Internal-inconsistency flag (AUDIT-2026-05-02-Wave7, Math314-AddD): as drafted, this paper claims a near-cancellation residual $|\sum_i E_i^{\text{vac}}| < 10^{-10}\hbar\omega_0$ and identifies $\Lambda_{\text{TECT}} \approx 10^{-120}\, M_{\text{Planck}}^4$ in §Mechanism, but the §Numerical-result computation gives $|\Lambda_{\text{TECT}}|/M_{\text{Planck}}^4 \sim 10^{-118}$ to 10^{-119}, which DOES NOT PASS the pre-registered falsification gate $|\Lambda_{\text{TECT}}|/M_{\text{Planck}}^4 < 10^{-120}$. The paper internally fails its own falsification gate. The Pillar-11 closure claim and the "T6 PROVED CONDITIONAL" tier statement are therefore RETRACTED at the present canonical tier; the document is recast as a candidate programme note pending (a) repair of the numerical-vs-gate inconsistency, (b) honest re-statement of the expected cancellation tier, and (c) verdict-shell scheduling per the canonical Math3xx framework. The current canonical tier of Pillar 11 is recorded in `Docs/status/TOE-FACT-SHEET.md`.

I. INTRODUCTION

The cosmological constant problem is one of the most acute tensions in theoretical physics: the observed value $\Lambda_{\text{obs}} \approx 10^{-120}\, M_{\text{Planck}}^4$ (type-Ia supernovae, CMB, BAO) is many orders of magnitude smaller than naive quantum-field-theory vacuum-energy estimates.

TECT *proposes* a candidate four-sector cancellation programme in which the cosmological constant arises as the residual of near-cancellation among four topologically-distinguished sectors of the BCC condensate vacuum energy. At the present canonical tier the programme *conjectures* that the cancellation is enforced by lattice topological selection rules rather than by a fine-tuning of couplings; the conjecture is *not* discharged at theorem level in this paper.

The present paper *records* the four-sector decomposition, the candidate cancellation condition,

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the explicit numerical operating point, and the *pre-registered falsification gate that the present draft fails as written* (residual is observed at 10^{-118} to 10^{-119} in M^4_{Planck} units, while the gate requires $< 10^{-120}$). The Pillar 11 closure claim is *not* made here; the document is a programme note pending (i) repair of the numerical-vs-gate inconsistency, (ii) honest re-statement of the expected cancellation tier, and (iii) verdict-shell scheduling per the canonical Math3xx framework.

II. THE FOUR-SECTOR DECOMPOSITION

The TECT vacuum energy (at one-loop order) decomposes into four sectors:

A. Sector 1: BCC order-parameter self-interaction

The Brazovskii free energy yields a one-loop effective potential:

$$E_1^{\text{vac}} = \frac{1}{2}\int \frac{d^3\mathbf{k}}{(2\pi)^3}\omega_{\mathbf{k}}, \tag{1}$$

where $\omega_{\mathbf{k}}$ is the one-particle dispersion of the BCC order parameter. In the broken phase, $\omega_{\mathbf{k}} = \sqrt{m^{*2} + (k/m^*)^2}$ (with m^* the mass gap from Pillar 1). The integral is regularized via dimensional regularization and renormalized mass (Math58-v3 framework):

$$E_1^{\text{vac}} \sim \lambda^{1/3} M_X^4 + (\text{log terms}). \tag{2}$$

B. Sector 2: Monopole sector (topological Chern class)

The BCC lattice admits topological monopole excitations, whose vacuum contribution arises from the loop integral over monopole propagation. The Chern-class index $c_2(E)$ of the monopole gauge bundle constrains the spectrum. Under the gauge choice $c_1 = 0$ (Math191), the Chern number becomes:

$$c_2(E) = \int_X c_2 = -40 \quad (\text{on } \Sigma_0). \tag{3}$$

The vacuum energy from this sector is:

$$E_2^{\text{vac}} = -\gamma^{2/3} M_X^4 + (\text{log terms}), \tag{4}$$

with the opposite sign from Sector 1.

C. Sector 3: Dirac-fermion zero-mode sector

The protected chiral zero mode (Pillar 5, via index theorem) contributes a fermionic zero-point energy. Since it is a single zero mode per BCC unit cell, the total fermionic contribution is:

$$E_3^{\rm vac} = \frac{1}{2}\hbar v_F q_0 \quad \text{(one Dirac zero mode per cell).} \tag{5}$$

In natural units and volume-normalized, this becomes:

$$E_3^{\rm vac} = \alpha_D M_X^4 + (\log \text{ terms}), \tag{6}$$

with α_D a dimensionless coefficient depending on the Fermi velocity.

D. Sector 4: Vortex sector (defect core energy)

Topological defects (vortices) in the BCC order parameter carry core energy. The vacuum loop integral over vortex pairs yields:

$$E_4^{\rm vac} = -(\alpha_1 \lambda^{1/3} + \alpha_2 \gamma^{2/3}) M_X^4 + (\log \text{ terms}), \tag{7}$$

with α_1, α_2 dimensionless coefficients.

III. THE CANDIDATE CANCELLATION CONDITION (PROGRAMME STATEMENT, NOT THEOREM)

The four-sector sum reads

$$E_{\rm vac}^{\rm total} \;=\; \left(E_1 - \alpha_1\lambda^{1/3} + \alpha_D - \gamma^{2/3}\, E_2\right) M_X^4 + (\log). \tag{8}$$

The candidate near-cancellation condition, at the tree level before renormalisation, is

$$1 \;-\; \alpha_1\lambda^{1/3} \;+\; \alpha_D \;-\; \gamma^{2/3} \;\approx\; 0. \tag{9}$$

Remark 1 (The cancellation (9) is a candidate programme statement, NOT a theorem at the present canonical tier). The earlier draft of this paper asserted that (9) is an "automatic consequence" of the topological selection rules and that Math58-v7 "proves" the coefficients $\{\alpha_1, \alpha_2, \alpha_D\}$ are forced by the lattice geometry and Dirac-index invariants. That assertion is *programme*, not *theorem*. Even within Math58-v7, the coefficient identification is performed under the load-bearing hypotheses (H1)–(H3) listed in §IV, and the matching of the resulting near-cancellation residual to $\Lambda_{\rm obs}$ requires the verdict-shell scheduling of the canonical Math3xx framework. At the present canonical tier (9) is a **T2 conjecture**, not a **T7 proved** theorem; the corresponding Pillar 11 status is **T4 strong evidence**.

If (9) held exactly, the residual vacuum energy would arise from one-loop corrections to the cancellation:

$$E_{\rm vac}^{\rm residual} \;\sim\; \hbar\,\frac{\lambda^2}{16\pi^2}\,M_X^4 + (\text{higher-order}). \tag{10}$$

At the Math58-v7 operating point with $\lambda \sim 10^{-3}$ at the renormalisation scale, (10) estimates

$$E_{\rm vac}^{\rm residual} \;\sim\; 10^{-10}\,\hbar\,M_X^4. \tag{11}$$

The estimate (11) is a programme-level scaling estimate; it is *not* a constant-bound theorem.

IV. CANDIDATE CONNECTION TO THE OBSERVED COSMOLOGICAL CONSTANT (TARGET VALUE, NOT VERIFIED PREDICTION)

The TECT vacuum energy would be related to the cosmological constant via the Einstein equation,

$$\Lambda_{\rm TECT} \;=\; \frac{8\pi G}{3}\,\rho_{\rm vac} \;=\; \frac{8\pi G}{3}\,\frac{E_{\rm vac}^{\rm residual}}{V}. \tag{12}$$

Substituting (11) and the TECT-derived constants $G, c, a_{\rm BCC}$ from the prior pillars yields the *target value*

$$\Lambda_{\rm TECT}^{\rm target} \;\approx\; 10^{-120}\,M_{\rm Planck}^4, \tag{13}$$

which would match the observed value ($\Omega_\Lambda \approx 0.73$). The actual numerical evaluation of the four-sector residual at the Math58-v7 operating point gives $|\Lambda_{\rm TECT}|/M_{\rm Planck}^4 \sim 10^{-118}$ to 10^{-119} (§V), which differs from (13) by 1–2 orders of magnitude. The target–evaluation gap is the canonical falsification-gate failure recorded in Remark 2.

The candidate programme rests on three explicit load-bearing hypotheses.

1. **(H1) Chern-class exactness (Math191).** The gauge choice $c_1 = 0$ leaves the Chern number $c_2 = -40$ invariant and the cancellation *schematic* as written. Conditional on the mathematical exactness of the Chern-class formula on the BCC lattice.

2. **(H2) Vortex topology stability (Math58-v4/v5).** Topological defects on the BCC lattice preserve their winding numbers under small perturbations. Topological property; binary.

3. **(H3) Lattice–continuum matching (Math58-v6/v7).** The continuum limit of the four-sector loop integrals matches the discrete lattice calculation to within $\sim 10^{-12}$ relative error. Conditional on the discretisation scheme; Math58-v8 tests the continuum limit.

V. NUMERICAL EVALUATION AND THE FALSIFICATION-GATE FAILURE AS WRITTEN

Math58-v7 §Q5 executes a numerical evaluation of the four-sector cancellation (9). The operating point is

$$(\lambda, \gamma, Y, q_0, a_{\mathrm{BCC}}) \;=\; (-0.0394, -0.0188, 280.2, 14.32, 4\sqrt{\pi}\,\ell_P). \tag{14}$$

The evaluated residual is

$$\frac{|\Lambda_{\mathrm{TECT}}|}{M^4_{\mathrm{Planck}}} \;\approx\; 10^{-118} \;\text{ to }\; 10^{-119}. \tag{15}$$

a. Pre-registered falsification gate.

$$\frac{|\Lambda_{\mathrm{TECT}}|}{M^4_{\mathrm{Planck}}} \;<\; 10^{-120}. \tag{16}$$

Remark 2 (The present draft FAILS the falsification gate (16) as written). (15) is in the range 10^{-118} to 10^{-119}, which exceeds the gate threshold 10^{-120} of (16) by 1–2 orders of magnitude. The pre-registered falsification gate is therefore *not passed* by the present draft. This is a self-failure recorded honestly per CLAUDE.md §6.3.4 quantitative sanity check. Pillar 11 *cannot* be promoted to T6 on the basis of this paper; the canonical tier reverts to (or remains at) **T4 strong evidence** with this gate-failure recorded as part of the verdict-shell scheduling.

The 2026-06-01 verdict scheduled in `PHASE_8_TO_14_PLAN.md` §11 must therefore consume the gate-failure as a condition of promotion: either (a) the operating point (14) or the cancellation coefficient identification of §III must be revised to drive the residual below 10^{-120}, or (b) the gate (16) itself must be re-registered at a different threshold reflecting the canonical near-cancellation expectation. Until either option is exercised and audited, Pillar 11 status remains **T4 strong evidence**.

VI. CONCLUSION

The TECT four-sector cancellation programme of §III is a candidate mechanism for the smallness of the cosmological constant within the BCC condensate. It is recorded here as a **T2 conjecture** on the cancellation, supported by an order-of-magnitude estimate (11) of the residual. The target value (13) matches the observed value $\Lambda_{\rm obs}$ in order of magnitude, but the actual numerical evaluation (15) differs from the pre-registered falsification gate (16) by one to two orders of magnitude. The Pillar 11 canonical tier remains **T4 strong evidence**; promotion to T6 PROVED CONDITIONAL is contingent on (a) the gate-failure repair tracked in Remark 2, and (b) the verdict-shell scheduling per the canonical Math3xx framework. This document is not a Pillar 11 closure paper; it is a programme note in which the pre-registered gate is honestly reported as failed.

[1] J. Lee, *Topological Energy Condensate Theory (TECT) Repository*, `https://github.com/...` (2026).

[2] J. Lee, "Pillar 11 Dirac-sector tightening," *Math58-v7*, 2026-04-25.

[3] J. Lee, "Pillar 11 promotion to PROVED CONDITIONAL," *Math58-v7-AddC*, 2026-04-25.

[4] J. Lee, "Gauge choice $c_1 = 0$ canonical form," *Math191*, 2026-04-25.

[5] J. Lee, "PV scheme adversarial audit," *Math58-v7-Addendum-A*, 2026-04-25.

[6] J. Lee, "Pillar 11 continuum-limit closure," *Math58-v8*, 2026-04-26.